Harsh Navin Agarwal

Software Engineering | Full-Stack Systems | Applied AI
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EDUCATION

The Pennsylvania State University — University Park, PA
Bachelor of Science in Computer Science; Minors in Mathematics and Economics — *Expected May 2027*
GPA: 3.88/4.0

EXPERIENCE

Software Engineer (Freelance) — Remote
Unique Properties – Real Estate — *May 2026 – Present*

PMS Pro – Work & Performance Management Platform | *React, TypeScript, PostgreSQL, Supabase* 2026

- Built and launched, as the sole developer, a platform that assigns responsibilities, KPIs and daily tasks to **173 employees across 19 departments**, replacing spreadsheets and saving **70 managers an estimated 2–3 hours each per week**.
- Implemented authentication and role-based authorization controls, ensuring employees could only access data and perform actions permitted by their assigned roles and reporting hierarchy.
- Built a nightly PostgreSQL cron job that generates recurring tasks; a unique constraint makes it safe to re-run, and it catches up after up to 14 days of missed runs. KPI scores are calculated from task completion, weighted by priority.
- Built configurable multi-stage review workflows that HR can reorder and assign per department or review cycle, with the database rejecting invalid stage sequences.
- Migrated **828 performance goals** from legacy spreadsheets through a validated Excel import with dry-run previews and full rollback on failure.

DrawVault – Construction Drawing Management System | *React, Node.js, Express, SQLite, Cloudflare R2* 2026

- Built, as the sole developer, a drawing-management system used daily by **20+ architects and project staff**, with full revision history, instant server-side login revocation and **96 automated tests** in CI.
- Automated document hand-offs to external recipients: the system emails a generated PDF cover sheet and records each recipient's confirmation through a unique one-time link, saving the client **an estimated 5 hours per week**.

Math Peer Tutor & Grader — University Park, PA
Penn State Learning / Department of Mathematics — *August 2025 – Present*

- Teach Calculus II and Vector Analysis to **20+ students weekly**; evaluate **100+ weekly submissions** using departmental rubrics.

TECHNICAL PROJECTS

Photo Shopping – Secure AI Grocery Assistant — GitHub
Python, FastAPI, macOS Vision OCR, OAuth 2.1, MCP

- Built an assistant that turns photos of handwritten grocery lists (English, Hindi, or Hinglish) into price-compared carts across **Blinkit, Zepto, and Instamart**, running OCR on-device and re-checking only low-confidence lines with a hosted model.
- Designed a validation engine that blocks any cart violating the user's brand, substitution, or budget rules before it is written.

News Buddy – Automated AI News Pipeline — GitHub
Python, LangGraph, ChromaDB, FastMCP, Docker, OpenTelemetry

- Built a LangGraph pipeline that fetches, deduplicates, summarizes, and publishes daily AI news, with swappable LLM providers and rule-based checks that retry low-quality summaries.
- Compared **four model and prompt alternatives** against fixed pass/fail criteria and kept the existing model when none met the bar; traced every model call with OpenTelemetry.
- Added semantic search over the news archive with ChromaDB and a Docker-packaged MCP server that lets AI assistants query it read-only.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, HTML/CSS
Frameworks: React, Node.js, Express.js, FastAPI, LangGraph
Data & Cloud: PostgreSQL, SQLite, Supabase, ChromaDB, Cloudflare R2, Vercel, Railway
Engineering: Git, GitHub Actions, Docker, CI/CD, OpenTelemetry, OAuth, JWT, RBAC, RAG, MCP